

Task Management Application

Application API Endpoints

User & Role Management APIs

User APIs

- POST /api/v1/users – Create a new user
- GET /api/v1/users/{userId} – Get user details
- GET /api/v1/users – List all users
- PUT /api/v1/users/{userId} – Update user profile
- DELETE /api/v1/users/{userId} – Deactivate/Delete user

Role APIs

- POST /api/v1/roles – Create role
- GET /api/v1/roles – List all roles

User–Role Mapping

- POST /api/v1/users/{userId}/roles/{roleId} – Assign role to user
- DELETE /api/v1/users/{userId}/roles/{roleId} – Remove role from user
- GET /api/v1/users/{userId}/roles – Get roles of a user

Project Management APIs

Project APIs

- POST /api/v1/projects – Create a project
- GET /api/v1/projects/{projectId} – Get project details
- GET /api/v1/projects – List all projects
- PUT /api/v1/projects/{projectId} – Update project
- DELETE /api/v1/projects/{projectId} – Close/Delete project

Project Ownership

- GET /api/v1/users/{userId}/projects – Get projects owned by a user

Task & Category APIs

Tables involved: Task, Category, TaskCategory

Task APIs

- POST /api/v1/tasks – Create task
- GET /api/v1/tasks/{taskId} – Get task details
- GET /api/v1/tasks – List all tasks
- PUT /api/v1/tasks/{taskId} – Update task
- DELETE /api/v1/tasks/{taskId} – Delete task

Task Assignment & Filtering

- GET /api/v1/projects/{projectId}/tasks – Tasks under a project
- GET /api/v1/users/{userId}/tasks – Tasks assigned to a user
- GET /api/v1/tasks/status/{status} – Filter by status
- GET /api/v1/tasks/priority/{priority} – Filter by priority

Category APIs

- POST /api/v1/categories – Create category
- GET /api/v1/categories – List categories

Task–Category Mapping

- POST /api/v1/tasks/{taskId}/categories/{categoryId}
- DELETE /api/v1/tasks/{taskId}/categories/{categoryId}
- GET /api/v1/tasks/{taskId}/categories

Comment & Attachment APIs

Tables involved: Comment, Attachment

Comment APIs

- POST /api/v1/tasks/{taskId}/comments – Add comment
- GET /api/v1/tasks/{taskId}/comments – View task comments
- DELETE /api/v1/comments/{commentId} – Delete comment

Attachment APIs

- POST /api/v1/tasks/{taskId}/attachments – Upload attachment

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- GET /api/v1/tasks/{taskId}/attachments – List attachments
- DELETE /api/v1/attachments/{attachmentId} – Delete attachment

Notification APIs

- POST /api/v1/notifications – Create notification
- GET /api/v1/notifications/{notificationId} – Get notification
- GET /api/v1/users/{userId}/notifications – User notifications
- DELETE /api/v1/notifications/{notificationId} – Delete notification

Advanced Use (Optional)

- GET /api/v1/users/{userId}/notifications/unread
 - PUT /api/v1/notifications/{notificationId}/read
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High-Impact Reporting Endpoints

✔ 1. User Productivity Report (Most Important)

Endpoint

GET /api/v1/reports/users/productivity

Why these matters

This is a real corporate KPI—used by managers to evaluate employees.

Business Output

JSON

```
[
  {
    "userId": 1,
    "userName": "John Doe",
    "totalTasks": 12,
    "completedTasks": 8,
    "pendingTasks": 4,
    "completionRate": 66.67
  }
]
```

Tables Used

- User
- Task

What it evaluates

- Aggregation (COUNT, CASE)
 - Business calculation (%)
 - Clean DTO modeling
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✔ 2. Project Summary Dashboard

Endpoint

GET /api/v1/reports/projects/summary

Why these matters

This simulates a real dashboard used in companies to track project health.

Business Output

JSON

```
[
  {
    "projectId": 4,
    "projectName": "New Product Launch",
    "totalTasks": 20,
    "completedTasks": 10,
    "inProgressTasks": 6,
  }
]
```

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```
"pendingTasks": 4,  
"completionPercentage": 50  
}  
]
```

Tables Used

- Project
- Task

What it evaluates

- Multi-table joins
 - Grouping logic
 - Derived metrics (progress %)
 - Business thinking
-

✓ 3. Overdue Tasks Report

Endpoint

GET /api/v1/reports/tasks/overdue?days=7

Why these matters

This is critical in real systems—missed deadlines = business risk.

Business Output

JSON

```
[  
{  
  "taskId": 5,  
  "taskName": "Design Landing Page",  
  "dueDate": "2022-03-15",  
  "assignedUser": "Jane Smith",  
  "projectName": "Website Redesign",  
  "daysOverdue": 10  
}  
]
```

Tables Used

- Task
- User
- Project

Important for:

- Date handling
- Business rule implementation
- Multi-table joins
- Filtering logic